

Immunogold labeling of perforin and serine esterases in granulated metrial gland cells.

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Cytotoxic T lymphocytes (CTL) and natural killer (NK) cells are cytolytic lymphocytes known to produce a pore-forming protein, named perforin or cytolyisin, that lyses target cells by creating large pores on the target plasma membrane. Besides perforin, the granules of CTL and NK cells contain a family of serine esterases. Perforin has also been localized in granulated metrial gland (GMG) cells of the murine embryo implantation site by light microscopic immunostaining. Ultrastructural immunogold labeling with antibodies against perforin and a serine esterase (MTSP 1 or granzyme A) shows that GMG cells contain both perforin and serine esterases in the fine granular matrix of their granules. Perforin has been located in all of the granules, whereas gold particles corresponding to serine esterases have been found in most of the granules. Results from the double immunogold technique indicate that perforin and serine esterases colocalize to most of the same granules in GMG cells. This study supports the view that GMG cells are related to cytolytic lymphocytes.

Differential expression of perforin in cytotoxic lymphocyte in HIV/AIDS patients of China.

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Cytotoxic lymphocytes are critical in the control of HIV replication, it has been shown that perforin is the key effector of killing machinery for CTLs and NK cells, so we investigated the circulating levels of perforin in CD8+ T cells and NK cells by flow cytometry intracellular stain in Chinese HIV infected individuals, its association with disease progression was analyzed. Our results showed that NK cells express perforin more efficiently than CD8+ T cells, CD8+ T cells expressed perforin higher than that of healthy controls, but NK cells expressed lower perforin than that of healthy controls, both were not correlated with disease progression. but significantly associated with their numbers, anti-retrovirus therapy had no evident effects on peforin expression in CD8+ T cells, but enhanced perfrin expression in NK cells, perforin expression in CD8+ T cells and CD16+ NK cells correlate with CD4+ T cell counts significantly in HAART-treated group. Therefore, different mechanisms may be involved in regulating peripheral perforin expression in different cell types.

Localization and identification of granzymes A and B-expressing cells in normal human lymphoid tissue and peripheral blood.

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Cytoplasmic granules from activated natural killer (NK) and cytotoxic T lymphocytes (CTL) contain a pore-forming protein, perforin, and several homologous serine proteinases called granzymes. Expression of these proteins correlates with the cytolytic potential of cytotoxic lymphocytes. Using a panel of MoAbs specific for human granzyme A and B, respectively, expression of these proteinases in non-pathological lymphoid tissue and peripheral blood lymphocyte (PBL) subpopulations was investigated. Using immunohistochemistry and double stainings, the phenotype of granzyme-expressing cells in lymphoid tissue was investigated. Granzyme-positive cells were detected in all lymphoid tissues tested. No large differences in the number and distribution between granzyme A- and granzyme B-positive cells were observed. The highest number of positive cells was located in the red pulp of the spleen. Significant

numbers were detected in tonsil, lymph nodes, liver and thymus. Low numbers were present in the lamina propria of non-inflamed stomach, small intestine and colon. Phenotypic analysis and cell sorting showed that most of the granzyme-positive cells in lymphoid tissue and PBL consisted of CD3-CD16+CD56+ lymphocytes. Hardly any granzyme-positive CD3+CD8+ CTL were present in peripheral blood. The synthesis of granzyme A as well as B by both CD3+CD16+CD56+ and CD3+CD8+ cells in peripheral blood was increased upon IL-2 stimulation. These results indicate that in normal lymphoid tissue the predominant cytolytic cell population is formed by the NK cells, and activated CTL are rare.

Granzyme B and perforin lytic proteins are expressed in CD34+ peripheral blood progenitor cells mobilized by chemotherapy and granulocyte colony-stimulating factor.

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Granzyme B and perforin are cytoplasmic granule-associated proteins used by cytotoxic T lymphocytes and natural killer (NK) cells to kill their targets. However, granzyme B gene expression has also been detected in a non-cytotoxic hematopoietic murine multipotent stem cell line, FDCP-Mix. The objective of the present study was to investigate whether granzyme B and perforin could be expressed in human hematopoietic CD34+ cells and if present, discover what their physiologic relevance could be. The primitive CD34+ human cell line KG1a was investigated first and was found to express granzyme B and perforin. Highly purified hematopoietic stem/progenitor cells were then selected using the CD34 surface antigen as marker. Steady-state bone marrow (BM) CD34+ cells did not contain these proteins. Peripheral blood (PB) CD34+ cells, which had been induced to circulate, were also analyzed. After chemotherapy (CT) and granulocyte colony-stimulating factor (G-CSF) treatment, CD34+ cells strongly expressed mRNAs and proteins of granzyme B and perforin. In contrast, CD34+ cells mobilized by G-CSF alone were negative. Western blot analysis further showed that granzyme B and perforin proteins were identical in CD34+ cells and activated PBLs. Such proteins might be implicated in the highly efficient migration of CD34+ stem/progenitor cells from BM to PB after CT and G-CSF treatment. The cellular adhesion mechanisms involved in the BM homing of CD34+ cells are disrupted at least temporarily after CT. The Asp-ase proteolytic activity of granzyme B on extracellular matrix proteins could be used by progenitor cells for their rapid detachment from BM stromal cells and perforin might facilitate their migration across the endothelial cell barrier.

Granzyme B regulates antiviral CD8+ T cell responses.

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CTLs and NK cells use the perforin/granzyme cytotoxic pathway to kill virally infected cells and tumors. Human regulatory T cells also express functional granzymes and perforin and can induce autologous target cell death in vitro. Perforin-deficient mice die of excessive immune responses after viral challenges, implicating a potential role for this pathway in immune regulation. To further investigate the role of granzyme B in immune regulation in response to viral infections, we characterized the immune response in wild-type, granzyme B-deficient, and perforin-deficient mice infected with Sendai virus. Interestingly, granzyme B-deficient mice, and to a lesser extent perforin-deficient mice, exhibited a significant increase in the number of Ag-specific CD8(+) T cells in the lungs and draining lymph nodes of

virally infected animals. This increase was not the result of failure in viral clearance because viral titers in granzyme B-deficient mice were similar to wild-type mice and significantly less than perforin-deficient mice. Regulatory T cells from WT mice expressed high levels of granzyme B in response to infection, and depletion of regulatory T cells from these mice resulted in an increase in the number of Ag-specific CD8(+) T cells, similar to that observed in granzyme B-deficient mice. Furthermore, granzyme B-deficient regulatory T cells displayed defective suppression of CD8(+) T cell proliferation in vitro. Taken together, these results suggest a role for granzyme B in the regulatory T cell compartment in immune regulation to viral infections.

Perforin, a pore-forming protein detectable by monoclonal antibodies, is a functional marker for killer cells.

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Perforin is one of the important cytolytic factors in cytotoxic T lymphocytes (CTL) and natural killer (NK) cells. In this paper, we report rat mAbs against mouse perforin established by immunization with a recombinant mouse perforin fragment. These mAbs reacted with purified mouse perforin prepared from cytoplasmic granules of an NK-like cell line in ELISA and Western blot analysis. However, none of these mAbs blocked the hemolytic activity of mouse perforin or absorbed it when fixed in the solid phase. These results indicate that all of these mAbs react with denatured but not with native mouse perforin. By using a combination of the mAbs, we established a sandwich ELISA, for quantitating the cellular contents of perforin. These mAbs were also useful for immunohistochemical staining analysis, and perforin was detected in the cytoplasmic granules of CTL and NK cell lines. Perforin was also detected in a minor population of lymphocytes of the spleen, liver, and lymph node. In normal spleen cells of 5- to 8-week-old mice, 12-15% of asialo GM1+ cells and 7-21% of CD8+ T cells were perforin-positive, but CD4+ T cells, B cells, and macrophages were totally negative. These data clearly show that perforin is expressed in cells of a cytotoxic character in normal mice, in the same way as in primed mice.

The CML experience to elucidate the role of innate T-cells as effectors in the control of residual cancer cells and as potential targets for cancer therapy.

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Considering the general view that unconventional immune effectors play a major role in antitumor immunity, we recently postulated that the distinct new innate CD8 T-cell pool (co-expressing the transcription factor Eomesodermin and innate markers such as KIR/NKG2A) may counteract tumor cells, and thereby be potential target for cancer therapy. Here, to test this assumption, we used successfully targeted anti-leukemic therapy discontinuation (TFR) in chronic myeloid leukemia (CML). Numerical and functional status of innate CD8 T-cells, iNKT cells and $\gamma\delta$ T-cells, in comparison with NK cells, was compared longitudinally between non-relapsed patients (i.e., with > 12 months TFR) and relapsed patients (i.e., who experienced molecular recurrence during the first 12 months after TKI cessation) in a prospective pilot cohort (n=32), starting from treatment discontinuation (D0). Perforin, a key cytotoxic immune player, was expressed in a significantly higher proportion of both innate CD8 T-cell and NK-cell subsets in non-relapsed patients, compared with relapsed patients at D0. In parallel, we assessed the expression of PD-1, an exhaustion marker used as target in cancer therapy. For all T-cell subsets,

surface-expression level of PD-1 decreased in non-relapsed patients compared with relapsed patients at D0. This was particularly the case when considering iNKT cells for which surface-expression level of PD-1 even decreased relative to healthy control subjects. Lastly, we found a negative correlation between the proportion of innate CD8 T-cells expressing PD-1 and those expressing perforin in non-relapsed patients at D0. The fact that this was not the case in conventional CD8 T-cells is compatible with a reprogrammed effector profile preferentially targeting innate CD8 T-cells in non-relapsed patients. All in all, our results highlight NK cells and innate CD8 T-cells harboring cytotoxic content, as well as global downregulation of PD-1-expression on effector T-cells, as potential predictive functional signatures for successful TFR in CML. Considering innate CD8 T-cells, further investigations are needed to determine whether their possible contributory role in cancer surveillance in CML could be extended to other cancers, and also whether their targeting by immune check-point inhibitors could enhance their anti-tumoral functions.

CD8+ T Cell Biology in Cytokine Storm Syndromes.

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Familial forms of hemophagocytic lymphohistiocytosis (HLH) are caused by loss-of-function mutations in genes encoding perforin as well as those required for release of perforin-containing cytotoxic granule constituent. Perforin is expressed by subsets of CD8+ T cells and NK cells, representing lymphocytes that share mechanism of target cell killing yet display distinct modes of target cell recognition. Here, we highlight recent findings concerning the genetics of familial HLH that implicate CD8+ T cells in the pathogenesis of HLH and discuss mechanistic insights from animal models as well as patients that reveal how CD8+ T cells may contribute to or drive disease, at least in part through release of IFN- γ . Intriguingly, CD8+ T cells and NK cells may act differentially in severe hyperinflammatory diseases such as HLH. We also discuss how CD8+ T cells may promote or drive pathology in other cytokine release syndromes (CSS). Moreover, we review the molecular mechanisms underpinning CD8+ T cell-mediated lymphocyte cytotoxicity, key to the development of familial HLH. Together, recent insights to the pathophysiology of CSS in general and HLH in particular are providing promising new therapeutic targets.

Human Granzyme K Is a Feature of Innate T Cells in Blood, Tissues, and Tumors, Responding to Cytokines Rather than TCR Stimulation.

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NK cells and CD8 T cells use cytotoxic molecules to kill virally infected and tumor cell targets. While perforin and granzyme B (GzmB) are the most commonly studied lytic molecules, less is known about granzyme K (GzmK). However, this granzyme has been recently associated with improved prognosis in solid tumors. In this study, we show that, in humans, GzmK is predominantly expressed by innate-like lymphocytes, as well as a newly identified population of GzmK+CD8+ non-mucosal-associated invariant T cells with innate-like characteristics. We found that GzmK+ T cells are KLRG1+EOMES+IL-7R+CD62L-Tcf7int, suggesting that they are central memory T and effector memory T cells. Furthermore, GzmK+ cells are absent/low in cord blood, suggesting that GzmK is upregulated with immune experience. Surprisingly, GzmK+ cells respond to cytokine stimuli alone, whereas TCR stimulation downregulates GzmK expression, coinciding with GzmB upregulation. GzmK+ cells have reduced IFN- γ production compared with GzmB+ cells in each T cell lineage. Collectively, this suggests

that GzmK+ cells are not naive, and they may be an intermediate memory-like or preterminally differentiated population. GzmK+ cells are enriched in nonlymphoid tissues such as the liver and adipose. In colorectal cancer, GzmK+ cells are enriched in the tumor and can produce IFN- γ , but GzmK+ expression is mutually exclusive with IL-17a production. Thus, in humans, GzmK+ cells are innate memory-like cells that respond to cytokine stimulation alone and may be important effector cells in the tumor.

Elevated Granzyme B in Cytotoxic Lymphocytes is a Signature of Immune Activation in Hemophagocytic Lymphohistiocytosis.

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Patients with hemophagocytic lymphohistiocytosis (HLH) exhibit immune hyper-activation as a consequence of genetic defects in secretory granule proteins of cytotoxic T lymphocytes (CTL) and natural killer (NK) cells. Murine models of HLH demonstrate significant activation of CTL as central to the disease pathogenesis, but evaluation of CTL and NK activation in children with HLH or inflammatory conditions is not well described. CD8 T cells only express granzyme B (GrB) following stimulation and differentiation into CTL; therefore, we reasoned that GrB expression may serve as a signature of CTL activation. It is unknown whether human NK cells are similarly activated in vivo, as marked by increased granule proteins. Perforin and GrB levels are measured in both CTL and NK cells by flow cytometry to diagnose perforin deficiency. We retrospectively compared GrB expression in blood samples from 130 children with clinically suspected and/or genetically defined HLH to age-matched controls. As predicted, CD8 expressing GrB cells were increased in HLH, regardless of genetic etiology. Remarkably, the GrB protein content also increased in NK cells in patients with HLH and decreased following immunosuppressive therapy. This suggests that in vivo activation of NK cells occurs in hyper-inflammatory conditions. We conclude that increased detection of GrB in CTL and NK are an immune signature for lymphocyte activation in HLH, irrespective of genetic subtype and may also be a useful measure of immune activation in other related conditions.